

WRITTEN BY

**Dipl.-Ing.****Daniel Ismail Mir**

is Research Assistant at the Institute of Automotive Engineering (IFS) of the University of Stuttgart.

**Dr.-Ing. Qirui Yang**

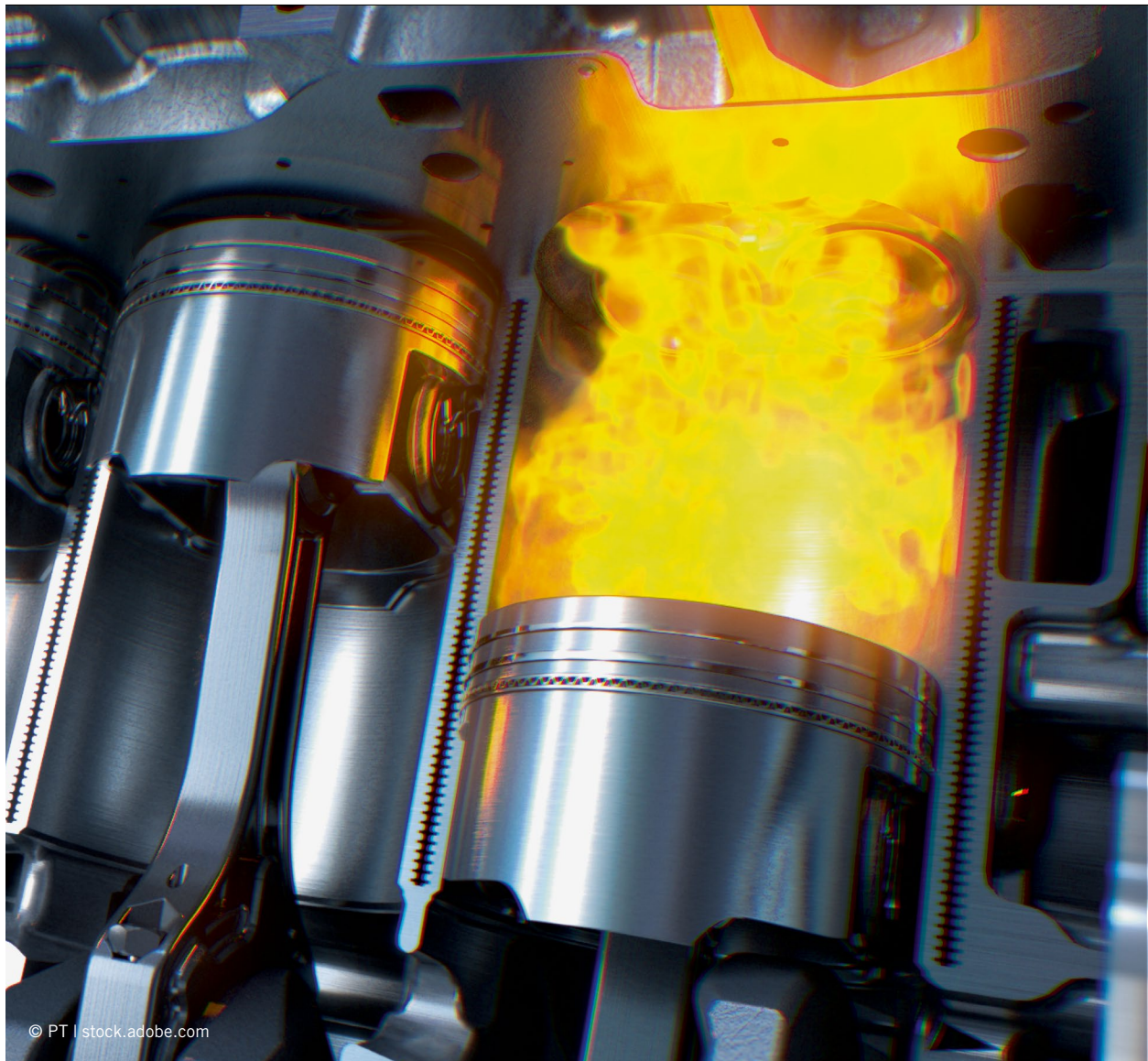
is Group Leader in the field of Artificial Intelligence and Simulation at the Research Institute for Automotive Engineering and Powertrain Systems Stuttgart (FKFS).

**Prof. Dr.-Ing.****André Casal Kulzer**

is Chair of Automotive Powertrain Systems at the Institute of Automotive Engineering (IFS) of the University of Stuttgart.

Prediction of Inhomogeneous Hydrogen Combustion with Direct Injection

The hydrogen engine plays a key role in the defossilization of the transport sector. Direct injection enables high specific power output, while the unique fuel properties of hydrogen allow efficient homogeneous lean combustion and minimum nitrogen oxide emissions. However, this also results in mixture inhomogeneities, which in turn may promote nitrogen oxide emissions. Within the FVV research project (No. 1454), this influence was investigated at the University of Stuttgart and incorporated into 0-D/1-D modeling.



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1 MOTIVATION

Spark ignition (SI) hydrogen combustion engines offer great potential for CO₂ emission reduction in the near future. Existing technologies can be utilized or easily retrofitted. Hydrogen (H₂) has advantageous combustion properties, in particular wide ignition limits and high laminar flame speeds. This enables homogeneous lean operation with low nitrogen oxide (NO_x) emissions and high thermal efficiency. Direct injection (DI) reduces the risk of combustion anomalies and increases power density. The injection timing plays a decisive role here. By injecting after the intake valve closes, the high displacement effect of H₂ can be avoided and the maximum power output potential can be achieved. However, this leads to mixture inhomogeneities which may affect the combus-

tion and nitrogen monoxide (NO) formation (as the main component of NO_x). This should be considered in the early design and development phase of an engine, in which 0-D/1-D simulations are usually employed. These simulation methods are known for short calculation times and yet high predictive capabilities. However, mixture inhomogeneities are usually not taken into account in this model class. Therefore, within the framework of the FVV project no. 1454 "Prediction of Inhomogeneous H₂-SI Combustion", the influence of inhomogeneities on combustion and NO formation was investigated at the Institute of Automotive Engineering (IFS) at the University of Stuttgart and integrated into 0-D/1-D simulation.

2 EXPERIMENTAL SETUP AND METHODOLOGY

In the project, experimental measurement data and 3-D simulation data based on Computational Fluid Dynamics (CFD) were analyzed. These originate from the FVV project no. 1449 "Near-zero Emission Concept for H₂ DI Otto engines." The test rig is operated at the Chair of Thermodynamics of Mobile Energy Conversion Systems (tme) at RWTH Aachen University. It is a modern long-stroke single-cylinder research engine with DI, characterized by high charge motion and a high turbulence level. The focus of the investigations was on lean operation ($\lambda = 2$).

The findings were incorporated into 0-D/1-D modeling. The FVV cylinder module was used as simulation environment. It was developed in an FVV research project [1] and is suitable for the overall process analysis, as numerous phenomenological sub-models such as for charge motion, turbulence and combustion are integrated.

3 MODELING OF MIXTURE HOMOGENIZATION

To predict inhomogeneities, the mixture formation process must be considered in 0-D/1-D simulation. For this, 3-D CFD

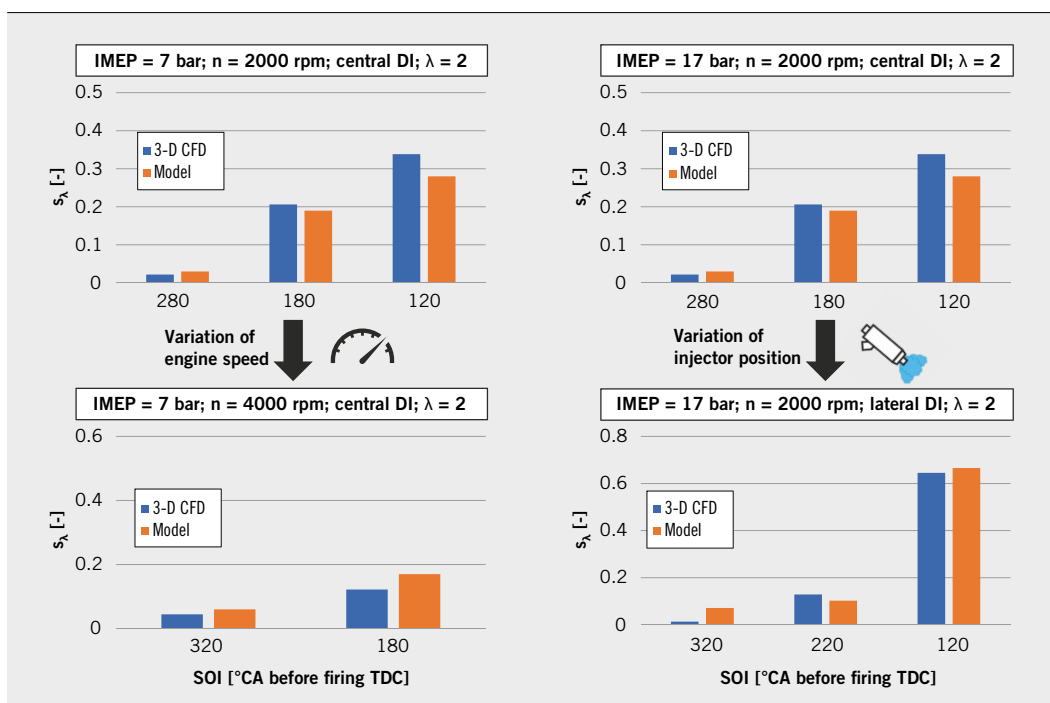


FIGURE 1 Comparison between 3-D CFD simulation and homogenization model with regard to the inhomogeneity degree s_k at ignition timing for different SOI at two different loads IMEP (top left and right); for variation of engine speed n (bottom left) and for variation of the injector position (bottom right) (© IFS)

simulations with varying injection timing were analyzed. In the modeling, the goal was not to compute the entire mixing process with highest accuracy including the complex injection process. Rather, it was aimed for predicting the degree of inhomogeneity at the start of combustion. The decisive factors identified were the injection time frame (particularly the start and end of injection) and the main flow in the combustion chamber (tumble flow). The homogenization process is enhanced by the high diffusivity of H_2 . The injection process plays a secondary role for the mixture field at ignition timing. As expected, late injection results in a higher degree of inhomogeneity.

The mixing process is simulated using a phenomenological homogenization model developed in an FVV research project [2]. It was originally elaborated for gasoline engines and is based on a two-dimensional finite volume approximation of the convection-diffusion equation in a representative sectional plane in the center of the cylinder, Eq. 1:

$$\text{Eq. 1} \quad \frac{\partial c}{\partial t} = \vec{\nabla} \cdot (D \vec{\nabla} c) - \vec{\nabla} \cdot (c \vec{u}) + \dot{q}_{DI}$$

This equation describes the temporal change in the spatial distribution of the passive scalar c , representing fuel and air. The velocity field $\vec{u}$ is approximated as a superposition of the piston movement and the tumble flow. The latter is modeled as an ideal flow field using Taylor-Green vortices. The diffusion term, with diffusion coefficient D , accounts for turbulent mixing effects. The H_2 DI is modeled through a source term $\dot{q}_{DI}$. After calibration for H_2 using a 3-D CFD simulation, the model is able to predict the global degree of inhomogeneity in the combustion chamber at ignition timing – defined as the standard deviation of the air ratio (s_λ). It should be noted that this model class is not able to predict the local mixture distribution. This would require more complex model approaches of a higher order, such as 3-D CFD simulations.

FIGURE 1 (top) shows the comparison between the 3-D CFD simulation and the homogenization model for variations of the Start Of Injection (SOI) at two different load points. The inhomogeneity degree s_λ at ignition timing is shown. The model successfully

reproduces the trend associated with the injection timing. There is good agreement with 3-D CFD simulation. The model also provides good results for changes in engine speed and injector position (central to lateral), **FIGURE 1** (bottom).

4 MODELING OF INHOMOGENEITIES

With respect to emission prediction, the distribution of λ and the temperature in the burned mixture is modeled to account for inhomogeneities. The underlying approach originates from an FVV project [3], which builds upon [2] and originally addressed gasoline engines. As part of this project, the approach was adapted for application to hydrogen engines. 3-D CFD simulations show that the distribution of λ in the combustion chamber at ignition timing corresponds to a normal distribution. Therefore, the mixture distribution is described by a density function φ using the inhomogeneity degree s_λ and the global air ratio λ , Eq. 2:

$$\text{Eq. 2} \quad \varphi(\lambda_{local}) = \frac{1}{s_\lambda \sqrt{2\pi}} \cdot \exp\left(-\frac{1}{2} \left(\frac{\lambda_{local} - \lambda}{s_\lambda}\right)^2\right)$$

Here, λ_{local} represents the local air ratio. For the 0-D/1-D simulation, the normal distribution is discretized into seven support points, **FIGURE 2**. The λ distribution defined at ignition timing is used as a fixed reference for the combustion process and the emission formation. Further analysis showed that the temperature distribution in the burned mixture is mainly determined by mixture inhomogeneities. This can be explained by the adiabatic flame temperature (which depends on λ) and is taken into account in modeling. Each local mixture inhomogeneity is assigned a corresponding temperature difference from the mean combustion temperature. This allocation and the resulting λ /temperature distribution are also shown in **FIGURE 2**.

5 MODELING OF COMBUSTION AND NO FORMATION

The collected measurement data show no significant influence of mixture inhomogeneities on the combustion for the research

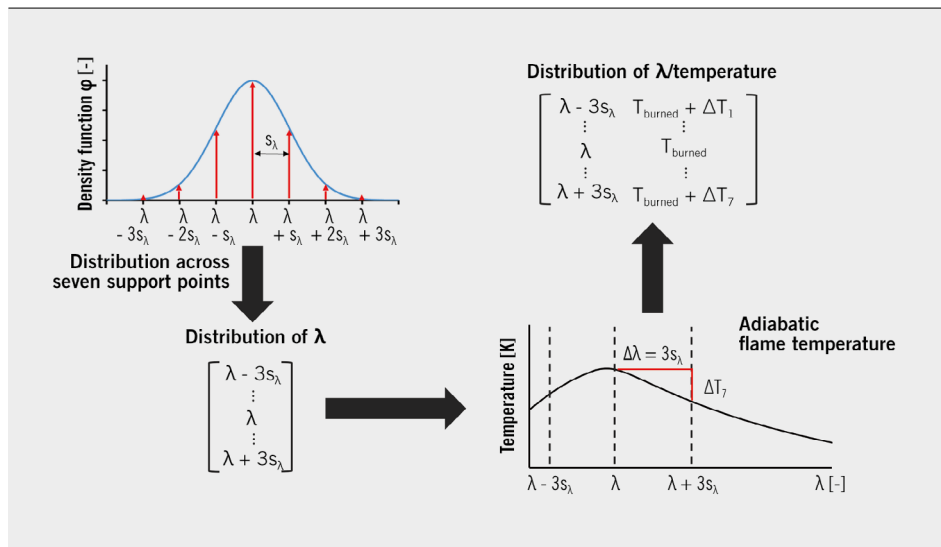


FIGURE 2 λ /temperature allocation: the normal distribution is discretized into seven support points (left); each support point of the λ distribution is assigned a temperature difference based on the adiabatic flame temperature (bottom left to top right) (© IFS)

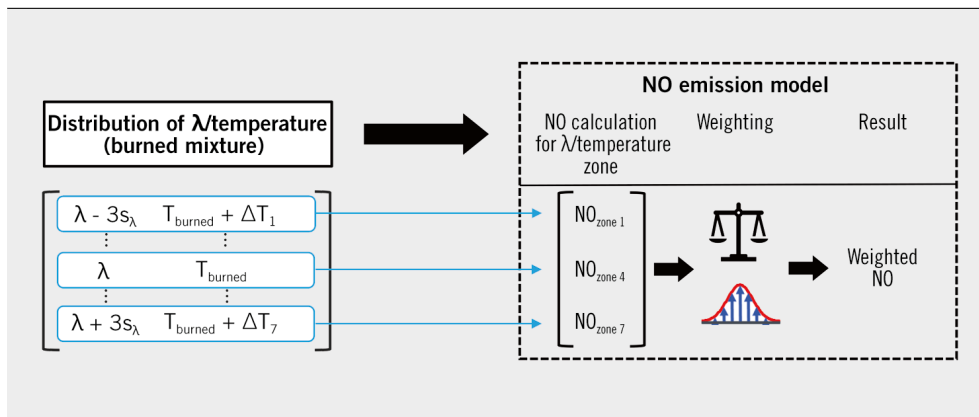


FIGURE 3 Calculation approach for NO emissions taking into account mixture and temperature inhomogeneities (© IFS)

engine under investigation. The reason for this is the high turbulence level during combustion. The combustion process can be accurately predicted using the well-established two-zone entrainment model for spark ignition engines, which assumes perfect mixture homogenization. Sub-models for caloric, laminar flame speed and auto-ignition behavior were adapted for H₂ combustion.

However, measurements and 3-D CFD simulations reveal a strong influence of mixture inhomogeneities on NO formation. A high inhomogeneity degree leads to an increase in NO in lean operation, as locally more fuel-rich zones with higher combustion temperatures promote NO formation. To account for this, a conventional NO model based on the extended Zeldovich mechanism is coupled with the previously defined λ /temperature distribution, **FIGURE 3**. For each λ /temperature zone, the NO value is calculated. Based on the normal distribution, all NO values are weighted and then combined so that a weighted NO value is the result. This approach significantly improves NO prediction. **FIGURE 4** shows the comparison between measurement data and 0-D/1-D simulations (without and with consideration of inhomogeneities). The simulation can reproduce the measurement trend very well. The error resulting from the assumption of a homogeneous mixture is significantly reduced by taking mixture and temperature inhomogeneities into account.

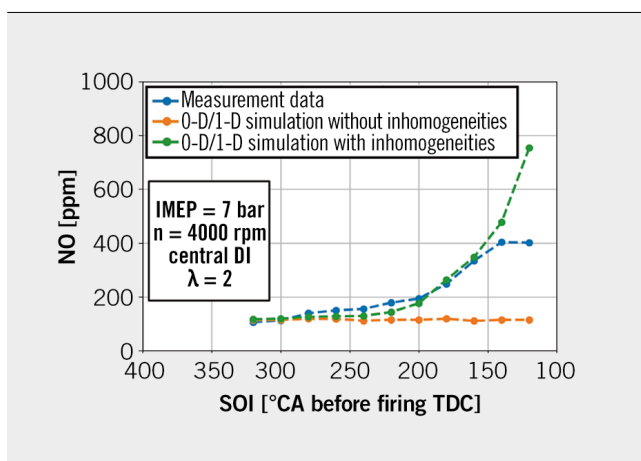


FIGURE 4 Comparison between measurement and 0-D/1-D simulations with regard to NO emissions for a variation of the SOI (© IFS)

6 SUMMARY AND OUTLOOK

Within the framework of the FVV project, the formation of inhomogeneities and their influence on combustion and the formation of NO emissions for a DI hydrogen engine were investigated and integrated into 0-D/1-D simulation. The measurements did not show any significant influence on the combustion process, whereas the NO emissions are strongly dependent on the inhomogeneity degree. The homogenization model presented takes into account the main factors in the mixing of H₂ and air and predicts the degree of inhomogeneity at ignition timing with high accuracy. Based on the inhomogeneity degree, a mixture distribution is defined. A corresponding temperature field is assigned to this distribution. Based on this distribution, NO emissions are calculated, which results in a significantly improved prediction. This provides added value in the development of injection strategies, where a trade-off exists between specific power output and NO_x emissions. The presented models should be validated on additional engines in the future.

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